# A PROCESS MODEL OF ENTREPRENEURIAL VENTURE CREATION

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**EXECUTIVE SUMMARY**

*The process model of entrepreneurial venture creation developed in this paper is based on interviews with entrepreneurs who started twenty-seven businesses in a range of industries in upstate New York. The venture creation process described here is an iterative, nonlinear, feedback-driven, conceptual, and physical process.*

*The model includes internally and externally stimulated opportunity recognition, commitment to physical creation, set-up of production technology, organization creation, product creation, linking with markets, and customer feedback. For analytical convenience, the process has been divided into the opportunity stage, the technology set-up and organization–creation stage, and the exchange stage. Business concept, production technology, and product are respectively the* core *variables representing the three stages.*

*Entrepreneurs introduce differing amounts of novelty at each core variable during venture creation, and the varying amounts of novelty qualitatively distinguish one kind of entrepreneurship from another.*

*For the researcher, the model suggests a better method for specifying samples of entrepreneurial firms. It shows how studies on the context of venture creation can be more specific, and proposes that novelty at the core variables be operationalized as a step toward defining the entrepreneurial content of ventures.*

*For the prospective entrepreneur, the model will serve as a useful road map. It will alert the entrepreneur to the strategic issues at each stage in the venture creation process, particularly when introducing significant novelty at any of the core variables.*

## INTRODUCTION

In the growing literature on entrepreneurship, there exist few empirical studies exploring and identifying the conceptually significant categories and subprocesses in venture creation.

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Journal of Business Venturing 9, 223–242
0883-9026/94/$7.00

Venture creation is the process that roughly begins with the idea for a business and culminates when the products or services based upon it are sold to customers in the market.

Descriptive accounts of venture creation are available as case studies in numerous texts (Timmons 1990). Sequences in venture creation presented as flow models exist at various levels of detail (Gartner 1985; Moore 1986; Vesper 1980; Webster 1976; Webster 1977). Life cycle views of ventures' growth and decline are also available (Kazanjian 1988; Kazanjian and Drazin 1990; Churchill and Lewis 1983; Kimberly and Miles 1980). And more recently, network models of venture creation have been proposed (Larson and Starr 1993). The categories and variables in the literature are often broad, apply to all ventures, and reveal no nuances among individual ventures. Others are excessively detailed and apply only to ventures in particular fields (VanderWerf 1993; Vesper 1980). Data-based process models of the steps leading to the birth of new ventures, however, are absent in the literature (Van de Ven 1992).

It would seem that the literature in mainstream economics, where the firm is a common unit for analysis, would offer insights into the processes of the birth of firms (Leff 1979; Leibenstein 1968; Baumol 1968; Evans 1949). Unfortunately, economics literature also fails in this respect. First, economists are not generally concerned with individual firms; the firm in the theory of the firm is a representative firm from an aggregate rather than any individual firm (Penrose 1980). Second, though recognized as important, entrepreneurship is poorly integrated into mainstream economics theory (Casson 1982), and few economists save those of the Austrian school emphasize its study. In sum, both researchers of organizations and entrepreneurship (Bull and Willard 1993; Kirchhoff 1991; Bygrave 1989a,1989b; Katz and Gartner 1988; Kimberly 1979; Low and Macmillan 1988) and economists (Loasby 1982; Baumol 1968; Buchanan 1979, 1982; Kirzner 1973, 1979; Nelson and Winter 1982) agree that the processes of the birth of firms need better understanding.

This study aims to generate a grounded–theoretical, integrative process model of entrepreneurial firm creation by linking conceptual categories and subprocesses in the firm creation process identified from interview data (Glazer and Strauss 1967; Yin 1989).

## AN ENCOMPASSING PROCESS VIEW—ITS NEED AND ABSENCE

Process studies can fulfill several needs at the present stage of entrepreneurship research.

### Provide an Integrative Framework

Firms vary vastly in their characteristics as do the entrepreneurs who create them (Gartner 1985). On one hand, studies of entrepreneurship must generate generalizable conclusions about variables relevant to all new firms. On the other hand, each business is conceived in extremely individualistic and personal ways, with myriad circumstances facing the entrepreneurs at start-up. Further, several disciplines study entrepreneurship from respective viewpoints. As a result, there is a bewildering variety of research on the subject. Organizing this variety in order to draw generalizable conclusions is a thorny, unresolved, and central problem of entrepreneurship theory (Cheah 1990; Wortman 1987; Bygrave 1989a, 1989b; Low and MacMillan 1988). The process model developed here provides an *integrative* framework to bring cohesion to this vast body of existing literature. And to the extent this model successfully integrates disparate bodies of research, it contributes to the advancement of entrepreneurship theory (Bacharach 1989; Whetten 1989).

## Integrate Subprocesses

The model will also help to integrate certain subprocesses that have been well studied. For example, Timmons et al. (1987) have described *opportunity recognition* as an important process in entrepreneurship. Processes of new product introduction and responding to customer feedback are also relevant to entrepreneurship (Lambkin 1988; Lieberman and Montgomery 1988; Schoonhoven, Eisenhardt, and Lyman 1990; Maidique and Zirger 1985; Child 1972). An overarching process model is necessary to integrate and place these phenomena in the context of entrepreneurial firm creation.

## Facilitate Impact of Context Studies

According to Van de Ven, Hudson, and Schroeder (1984), the links between the context and critical elements in the venture creation process are not clearly known. This is because theoretical concepts in the process are not well established. If they were, it would be possible to explore specifically the impact of the context upon the various steps in the venture creation process (Dorfman 1983; Dunkelberg et al. 1987; Pennings 1982a, 1982b). At present, despite good correlations between contexts favoring firm creation and the actual emergence of firms, we have no *explanation* of the birth of firms (Van de Ven 1993; Bollinger, Hope, and Utterback 1983; Calvin 1984). This paper identifies new conceptual categories and subprocesses in firm creation, and thereby facilitates studies of the impact of context upon it.

## Facilitate Comparison with Internal Corporate Venturing (ICV)

Comparisons between internal corporate venturing (ICV) (Kanter et al. 1990; Pinchot 1985; Fast 1979) and entrepreneurial firm creation processes are not possible because there are no well-grounded process models of entrepreneurial firm creation. If process models in entrepreneurship could be compared to process models of ICV (Burgelman 1983; Hornsby et al. 1993), we would be able to distinguish the entrepreneurial elements from the functional elements in both processes. Theoretical and practical insights from such studies would be valuable for entrepreneurs, managers, and researchers.

## Facilitate Taxonomy Building

Research populations of entrepreneurial firms (and entrepreneurs) are often selected along convenient (e.g., industry, size, technology, or region) rather than taxonomic dimensions. Results from samples of firms chosen along convenient dimensions rarely extend to all firms along such dimensions (McKelvey 1982). Comparative process studies of new venture creation are necessary in order to define taxonomic populations of entrepreneurial firms (Hambrick 1984; Gartner, Mitchell, and Vesper 1989). This paper develops one of the first process studies helpful for taxonomy building.

Two reasons explain the *absence* of grounded process models of firm creation. First, process studies are time-consuming and difficult to conduct with limited resources, and interdisciplinary process studies are even more difficult than those within a well-defined field. Entrepreneurship studies, spread across academic boundaries, require the researcher to be familiar with multiple literatures, diverse methodological perspectives, and different units of analysis (Van de Ven 1992; Low and MacMillan 1988; Kilby 1971; Wortman 1987).

Second, at a simplistic and intuitive level, the stages in venture creation appear to be well identified, though only rarely is the existence of the stages substantiated by empirical evidence (Montanari et al. 1990; Venkataraman et al. 1990; Venkataraman 1989; Ruhnka and Young 1987). Further, case studies in entrepreneurship and ICV seemingly describe the processes of venture creation (Kanter 1983, 1989; Kidder 1981). Researchers therefore use cross-sectional methods to study parts of the entrepreneurial process (Kimberly 1979).

Cross-sectional entrepreneurial research has several limitations. First, the phenomenon under study cannot be placed in a holistic context. Second, the relative weight of the phenomenon in the overall process is not known. Third, unless we can generate taxonomic samples of entrepreneurial firms, cross-sectional results have only limited external validity (Cook and Campbell 1979). We know that only a fraction of new ventures use venture capital. In the absence of taxonomic samples, the generalizability of conclusions from cross-sectional studies about venture capital use is therefore suspect.

In sum, practical difficulties, the dominant received tradition of cross-sectional research, and the interdisciplinary nature of entrepreneurship all explain the absence and point out the need for empirical process studies of entrepreneurship.

## METHODOLOGY, RESPONDENTS, AND VALIDITY

Simon (1969) writes, "One should not let one's discipline determine the choice of method; rather, one should fit the method to the problem." Further, "entrepreneurship is one of the youngest paradigms in management sciences" (Bygrave 1989a, p.7), there is no general agreement on the defining concepts and variables characterizing it, and the processes of the birth of firms are not well understood (Low and MacMillan 1988). Given this situation, an exploratory field study involving interviews, "mapping systems" according to Simon (1969), was undertaken to obtain an overview of venture creation processes.

All respondents were selected from a city in upstate New York. They represented both genders, various professions, different social classes, and different age groups. The city has a diverse economic base, uninfluenced by the presence of any one industry or company. The businesses selected comprise the categories in Table 1. The aim in selecting particular businesses was to cover a variety, both in manufacturing and services, in high-technology–based fields as well as the more traditional areas. The set is not a random sample, but given that taxonomic samples of entrepreneurial firms cannot be identified at the present (Gartner, Mitchell, and Vesper 1989; McKelvey 1982; Webster 1977), it did fairly represent entrepreneurial businesses in the region (Yin 1989).

The interviewing technique involved asking an open-ended question to introduce a topic (Payne 1951; Bradburn and Sudman 1979). This gave the respondents considerable leeway in giving answers. Based on respondents' answers, and if additional information was necessary, probing questions were asked. With successive interviews, themes and issues recurred within each business category and across all categories. By the twentieth interview, approximately five in each category, there was repetition of process-related detail. Additional interviews were carried out to ensure external validity. Given that interviewing is expensive and time-intensive, the number of interviews conducted is believed to be adequate for generalizations about the entrepreneurial process.

Once the model was developed, it was presented to a group of entrepreneurs in the region, most of whom had participated in the interviews, to test if it captured the complexity of the process and whether it matched their experiences (Yin 1989). Their feedback was incorporated into the model.

**TABLE 1.** Summary of Data

| Firms | Opportunity Recognition 1. | Business Concept 2. | Business Concept Development 3. | Production Technology 4. | Production Technology Development 5. | Organization Creation: Resource Need 6. | Product 7. | Product Development 8. |
|---|---|---|---|---|---|---|---|---|
| Trade and distribution businesses | | | | | | | | |
| A1. | I | L | N | L | N | H | L | N |
| A.2 | E | L | N | L | N | L | L | N |
| A.3 | E | L | N | L | N | M | L | N |
| A.4 | E | L | N | L | N | M | L | N |
| Financial services and management consulting services | | | | | | | | |
| B.1 | E | L | N | L | N | L | L | N |
| B.2 | E | L | N | L | N | L | L | N |
| B.3 | I | L | N | L | N | L | L | N |
| B.4 | I | H | Y | L | N | L | H | Y |
| B.5 | E | L | N | L | N | L | L | N |
| B.6 | E | L | N | L | N | L | L | N |
| B.7 | E | L | N | L | N | L | L | N |
| B.8 | E | L | N | L | N | L | L | N |
| Computer-based services | | | | | | | | |
| C.1 | E | L | N | L | N | L | L | N |
| C.2 | E | L | N | L | N | L | L | Y |
| C.3 | I | L | N | L | N | L | H | Y |
| C.4 | I | L | N | L | N | L | L | Y |
| C.5 | E | H | N | L | N | M | H | N |
| C.6 | I | L | N | L | N | L | H | Y |
| C.7 | E | L | N | L | N | L | H | Y |
| Electronics and technology-based design and manufacturing | | | | | | | | |
| D1. | I | L | N | L | N | H | L | Y |
| D.2 | I | L | N | L | N | L | H | N |
| D.3 | E | L | N | L | N | H | L | N |
| D.4 | I | L | N | L | N | L | H | Y |
| D.5[a] | I | L&H | Y | H&L | Y | H | L&H | Y |
| D.6 | E | L | N | L | N | H | L | Y |
| D.7 | E | L | N | L | N | L | L | Y |
| D.8 | I | L | N | H | Y | H | H | Y |

*1. Opportunity Recognition: Internally Stimulated (I) or Externally Stimulated (E).*
*2. Business Concept: High Novelty (H) or Low Novelty (L).*
*3. Business Concept Development: Yes (Y) or No (N).*
*4. Production Technology: High Novelty (H) or Low Novelty (L).*
*5. Production Technology Development: Yes (Y) or No (N).*
*6. Organization Creation: Resources Needed, High (H), Moderate (M) or Low (L).*
*7. Product: High Novelty (H) or Low Novelty (L).*
*8. Product Development: Yes (Y) or No (N).*
*[a] The entrepreneur introduced two distinct products.*

## Model Development

Numerous sequences in venture creation are available in the literature (Vesper 1980). For example, Gartner (1985), has stated that the entrepreneur locates a business opportunity, accumulates resources, markets products and services, produces the product, builds an organization, and responds to government and society. This broad description of the venture creation process, being applicable to all new ventures, served as the starting point for organizing data (Eisenhardt 1989; Bogdan and Bilken 1982).

The interview data were transcribed to generate case studies. Next, all quotes and facts from the transcripts that related to opportunity recognition, technology, customer, product, and so on, were listed together and closely examined for patterns. Assembling and analyzing was carried out for all known themes and variables in venture creation. Conceptual categories and subprocesses emerged inductively from this examination, and were then refined and linked to generate the model. No conscious, preconceived theory guided their identification (Eisenhardt 1989; Yin 1989; Burgelman 1983; Glazer and Strauss 1967; Simon 1969).

The method of drawing inferences and identifying conceptual categories parallels the methods used by Burgelman (1983) in studying internal corporate venturing. The resulting model based on grounded data offers an integrated view of firm creation starting from opportunity recognition through to the first sale and customer feedback.

In this paper, the word "firm" is used to denote a conceptual entity and "venture" or "business" to mean an actual business. Data about each venture are classified into conceptual categories in Table 1. The assigned values of "high" or "low" are adequate for first-order analysis but need to be refined in future research. To illustrate the issues, direct quotations are used wherever possible. Respondents are identified by their initials to maintain confidentiality. "Product" refers to both manufactured goods and services.

## SUBPROCESSES AND CONCEPTUAL CATEGORIES FROM DATA

When answers to the question, "Could you tell me how you began your business?" were assembled, it was apparent that diverse personal circumstances prompt venture creation. Upon closer examination, two distinct routes in the opportunity recognition process were discerned. They are described under "externally stimulated opportunity recognition" and "internally stimulated opportunity recognition," and are shown in Figure 1.

### Externally Stimulated Opportunity Recognition

The *decision to start* a venture preceded *opportunity recognition* for certain entrepreneurs. The decision was influenced by the entrepreneurs' personal and environmental circumstances at that time. For example, some entrepreneurs in the set decided to begin at particular stages in their personal lives, some when their employers decided to relocate and they did not wish to move, and some because they could not bear to work for others any longer. The reasons given were numerous and specific to the individuals at the time. But what all had in common was that the *decision to start* a venture preceded the choice of a particular opportunity for committed pursuit.

Here are IC's words describing how he started his specialty publishing and consulting venture:

> I had been in the same level in the company for five years. I was blocked off from the top job in my field, and at the same time, I was approaching my — birthday. I had been talking about it (starting my own business) for years. Here I am, reasonably in possession of my faculties, and so I thought this is as good a time as any.

Most entrepreneurs who decided to begin ventures in this manner recognized vastly more opportunities than they seriously chose to pursue. Their effort was to avoid getting distracted by these recognized opportunities. In HM's words:

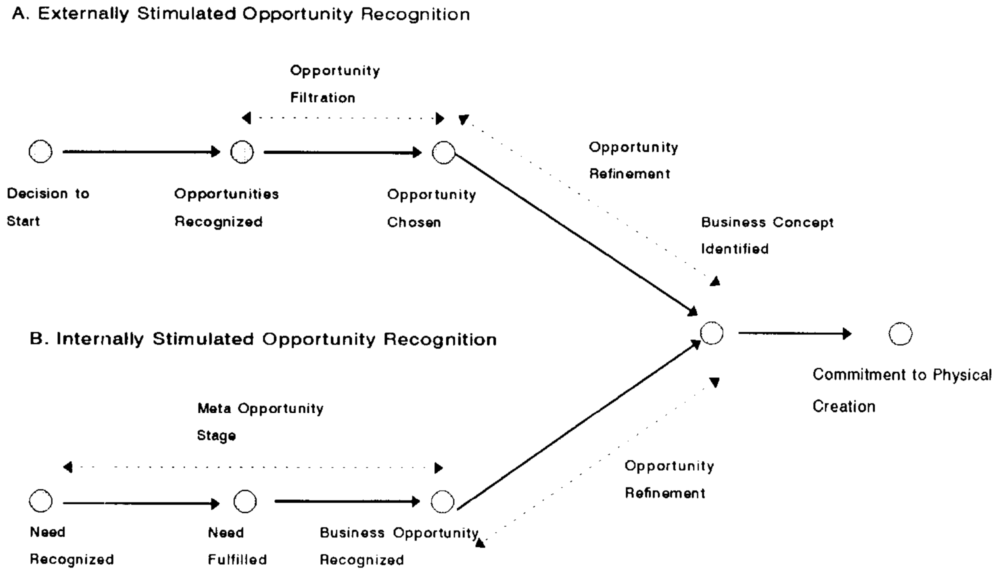


**FIGURE 1** Opportunity recognition sequences in entrepreneurial venture creation.

> ... there are so many opportunities out there, oh, so many ... every time an opportunity comes along, I think I'm supposed to take advantage of it. So I get lost, who knows where, until it becomes apparent I have no experience to know how to make it happen, no base of knowledge. So I see lots of opportunities every time I turn around, but I (tell myself) that's not where you belong ...

The decision to start was therefore followed by a search to align the prospective entrepreneurs' knowledge, experience, skills and other resources with market needs. In seeking this alignment, entrepreneurs eliminated inappropriate opportunities. In the end, only one, or a few, were chosen for committed pursuit. There was thus a *filtration of opportunities* from among a host of recognized ones (Long and McMullan 1984).

Once the *commitment to pursue* particular opportunities was made, entrepreneurs refined the opportunities. PA described this as the "massaging" of ideas: "an entrepreneur may see an idea and massage the idea until he makes it a good idea. If someone had looked at the idea previously, without the entrepreneurial glasses on, he may not see it in the same vein." The massaging, or elaboration, or *refinement* of opportunities chosen (Rockey 1986) resulted in the identification of *business concepts.*

Once entrepreneurs had focused on respective business concepts, they could easily articulate aspects of their ventures that distinguished them from those of others in the same field. To illustrate, in JM's words:

> the product we are handling is very up-scale in terms of price, quality and performance. We're not in the lower level where there's competition. Our niche is high up for sophisticated systems and exotic materials.

HM said:

> I believe that we are all unique and we've got our own little piece that will make it different than anyone else even though we look alike, sound alike, and walk like everyone else.

> There are a lot of common denominators, but there is one that separates us and that's one we capitalize on to make the difference.

In over half of the businesses (59 %), the decision to start a venture was followed by opportunity recognition and opportunity filtration. An opportunity suitable to the entrepreneur was chosen for committed pursuit. The chosen opportunity was refined to arrive at the business concept. Opportunities identified and pursued in this way, after an "opportunistic search" (Cyert and March 1963), are defined as "externally stimulated opportunities."

## Internally Stimulated Opportunity Recognition

For some entrepreneurs, *opportunity recognition* preceded the *decision to start* their ventures. The prospective entrepreneurs experienced, or were introduced to, needs that could not be easily fulfilled through available vendors or means. They tried, on their own or with others, to find their own solutions to satisfy the needs. At this stage, the entrepreneurs did not always realize that the needs they sought to fulfill were broad and that businesses could be created in fulfilling them. Because the business objectives were not clearly recognized yet, this stage is referred to as the *meta-opportunity stage.*

Once solutions to satisfy the needs were found, the business possibilities became apparent. Over time, the business possibilities became attractive and the *decision to start* ventures was made. This was again followed by *refinement of opportunities,* which led to the identification of *business concepts.*

SW started his violin repair business as follows:

> I couldn't find anyone I had enough faith in to repair violins I was playing, so I started repairing myself, and the word got around that I would do that. So I started doing that. After a while it got to be a burden to do it for free, and I started charging people for it.

GB's business in medical electronics had similar origins. A doctor friend casually expressed the need for certain kinds of measurements. Instruments to take those measurements did not exist. As a researcher in electronics, GB realized that an instrument could be built to fulfill this need, and took up the challenge to do so. A prototype was soon assembled in his laboratory. A refined version of the prototype became the basis of GB's multimillion dollar business.

A good many businesses (41%) in the set started in this way. A personal need was first identified and then fulfilled. The realization that the need was widespread led to the recognition of the business opportunity. The opportunity was refined to define the business concept. This was followed by a commitment to pursue the opportunity. Opportunities recognized and pursued after "problemistic search" (Cyert and March 1963) are defined as "internally stimulated opportunities."

## Business Concept and Business Concept Development

Both externally and internally stimulated opportunity recognition processes culminated in the identification of the business concept. Its identification is evidence of the successful completion of the subprocesses of *opportunity filtration, opportunity selection,* and *opportunity refinement* in venture creation as indicated in Figure 1.

Every venture had locational, product-market, or other features that were unique and gave the business competitive advantage. Every venture also was located at a certain point on an evolving industry structure and trajectory, and the dynamics of that industry structure created for the venture certain opportunity windows, advantages, and risks (Roure and Keeley 1990; Roure and Maidique 1986). At this point, the businesses were defined with sufficient focus so that the entrepreneurs could succinctly describe respective business concepts (Abell 1980).

When these statements were examined, it was evident that most ventures were based on business concepts familiar to respective customers. Only a small fraction of the businesses were based on truly novel business concepts. In Table 1, these are shown to have high business concept novelty while the rest have low business concept novelty.

Though a business concept may be defined well enough for an entrepreneur to start a new venture, it may not be congruent with or sufficiently matched with customer needs (Block and MacMillan 1985). This was true for two of the three businesses with high business concept novelty. They had to further develop their business concept (MacMillan, Zemann, and Narasimha 1987; Hofer and Sandberg 1987).

*Business concept development* refers to the effort involved in clarifying the business concept in order to achieve a good fit between customer needs and the entrepreneur's perceptions of those needs. When business concepts were novel, there were no precedents nor any customer feedback to guide the entrepreneurs. They had to introduce their products to customers, receive feedback, and only then develop further their business concept until a close match with customer needs was established (Maidique and Zirger 1985).

Business concept development was not an issue for entrepreneurs whose businesses had low business concept novelty. This was because other businesses existed that fulfilled similar customer needs, and the level of business concept resolution necessary for a good fit with customers was generally known. Entrepreneurs had only to achieve comparable or slightly different or better clarification of their business concepts.

High business concept novelty not only demanded the effort at business concept development, but also posed serious *marketing* challenges. JA's idea of selling product innovation consulting services was novel at the time he began. Convincing customers was extraordinarily difficult for him because customers were unused to contracting for such services. He had to overcome the resistance of organizational bureaucracies and their fear of trying out something new (Schumpeter 1934). He described his plight as follows:

> I thought these companies will be happy to find ways to . . . that they would need this kind of service. But the fact of the matter was that they weren't ready for it and didn't understand what it was. From my point of view, if I had gotten in the pizza business, it would have been easier. You know what that is. What I was doing was missionary work—education for the client.

Similarly, GB's product when first introduced was "too researchy" and failed from a "market point of view." His customers did not know what it was or how it worked. GB had to further develop his business concept, educate customers, and be educated by them before his product found customer acceptance. JA and GB had to *create* markets.

Most entrepreneurs in the set, however, *responded* to markets. In SB's words:

> The market essentially exists. I think that small companies by and large do not create markets. They are too small. By and large they have to find something within an existing area. Entrepreneurs don't have staying power. It requires a lot of money, a long time . . . creating a new market is an extremely difficult, treacherous, and expensive affair.

HM's business also had high business concept novelty, but the need for her product was well-recognized at that time. Fortunately for HM, there was congruence between customer needs and HM's perceptions of those needs. There was little need for customer education, and marketing did not pose a serious challenge. It was as if HM had found a timely and a novel cure for a long known illness. Though the business concept was novel, it did not require effort at business concept development or at market creation.

Whereas both JA and GB began their ventures from internally stimulated opportunity recognition, HM began from externally stimulated opportunity recognition. Future research will have to explore whether entrepreneurial effort at business concept development and marketing is greater when high business concept novelty is accompanied by internally stimulated opportunity recognition.

## Commitment to Physical Creation

The processes culminating in business concept identification, whether internally or externally stimulated, required varying amounts of time for different respondents. Further, most of this effort was intangible, and generally required entrepreneurs' personal resources, mostly time. As JW explained:

> For the better part of two years, I was busy developing ideas, getting myself organized, and ready to make a move. It wasn't an overnight decision . . . No question, a lot of thought went into it, and has to go into it . . . You don't do it quickly, or lightly, or without proper planning, or preparation.

To proceed beyond business concept identification, entrepreneurs required physical and other resources that were beyond most respondents' private means. An organization had to be created and production technology set up to transform the business concept into a marketable product. The decision to seek and invest outside resources to pursue the venture with commitment was a significant stage in the venture creation process. In DK's words:

> . . . that I can do this in a laboratory doesn't mean a squat; it's useless information. *Who's going to make a thousand tons of it and by what procedure?* How do you jump from a beaker [which is] a good idea you can demonstrate . . . Can you take it from this size and jump to that size? (emphasis added).

The really critical step according to PS is "when a person or a group of persons get up and strike out on their own . . . when they have accepted the fact that they are going to do it and accept all the pain that is going to occur." The *commitment to physical creation* is thus a significant transition point in venture creation.

## Production Technology and Production Technology Development

Production technology set-up and organization creation followed the commitment to physical creation. The principal technological apparatus for service businesses in the set was the expertise and knowledge of the entrepreneurs. Beyond basic office facilities and equipment, their resource needs at start-up were modest. Certain other businesses in the set required considerable resources for the set up of production technology. Sources of funding varied widely; many respondents used personal funds, some had venture capitalist backing, and at least one was partially funded by a parent company. Many skirted with bankruptcy for lack of funds, and a few had failed before in earlier ventures.

For all the visible variety and the differing resource needs, however, all businesses in the set except a few used standard equipment and technology to create respective products. Only the few businesses having "high" production technology novelty (Table 1) *developed* their technologies in their laboratories. Successful product creation, contingent upon the development of the underlying technology, introduced uncertainty, required risk capital, and made venture creation qualitatively more hazardous for these businesses.

Compared to the time devoted to discussing opportunity recognition, feedback from customers, environmental factors beyond entrepreneurs' control, or technological and market uncertainty, respondents hardly talked about organization creation and the set-up of production technology. Gathering resources and maintaining their availability—financial and other—was viewed as something to be taken care of, or obtained as a utility, rather than as an issue of strategic consequence. This is paradoxical and instructive given that the mechanics of resource gathering and organization creation is the very substance of texts and much research, and the most visible stage in venture creation. The only exceptions were the few cases where novel technology was being developed in the venture creation process. Once overcome, resource issues were narrated as humorous war stories. In DK's words: "We went to zero thrice in one year," or "Venture capitalists? Good people, but they gamble. There are no prophets in this business."

## Organization Creation

*Organization creation* occurred in parallel with the set-up of production technology for all ventures. Organization creation refers to the building of the *physical structure* as well as *organizational processes* that surround production technology at the core (Thompson 1967).

Over half the businesses (59%) were less than five years old at the time of interviews. Many had not stabilized operationally, and organizational structures and practices were at various stages of evolution. Departmental boundaries were fluid, functional differentiation was limited, and titles meant little. Work was distributed among partners as much along respective specializations as convenience. The entrepreneurs, as is proverbially described, wore many hats. Feedback from customers as well as learning led to formalization; what worked in the previous iteration was continued until change was indicated. With every iteration, additional resources were put into the venture for growth (MacMillan 1983).

In their formlessness and frenetic pace, the new organizations resembled "synthetic organizations" (Thompson 1967), that is, organizations created to provide relief during natural calamities such as famines or floods, and where little attention is paid to functional differentiation. But whereas synthetic organizations are disbanded when the emergency prompting their creation is over, the newly created ventures concentrated on generating new customers, creating better products, and growth.

## Product and Product Development

About one-third of the businesses introduced "high" novelty products to customers (Table 1). High product novelty was generally accompanied by *product development.* In certain computer-based services and electronics-based ventures, there was product development even for low novelty products due to *customization.* Just as in business concept or production technology development, product development consumed resources, often delayed product introduction, and added to uncertainty and risk.

Entrepreneurs viewed products in their relationship to customers and markets, and as embodiments and subsets of business concepts. The product was, as it were, a marketable form of the business concept, to be changed as dictated by customer needs. Sometimes customer needs changed rapidly, driven by rapid technical change in certain industries. The product (and sometimes the business concept) had to be equally rapidly modified to satisfy changing customer needs. Entrepreneurs in such industries had to constantly and feverishly evaluate their business concepts in relationship to current markets. In stable industries, the evaluation of business concepts and products with respect to markets was a low-key activity for entrepreneurs (Sandberg and Hofer 1987; Meyer and Roberts 1986).

The evaluation carried out by the entrepreneur between the business concept and product on one hand, and customers and markets on the other, suggests the existence of a boundary between the entrepreneurial firm and its customers. The entrepreneur and the new venture belong on the supply side of the boundary whereas the market lies on the demand side.

## The Supply and Demand Boundary

Chronologically or linearly viewed, marketing the product across the supply and demand boundary is the next step in venture creation. The reality for most respondents was not quite so linear. Although individual arrangements varied, most had their initial customers lined up well before product creation. It was only when there was a customer, or a fairly guaranteed potential customer, that the physical creation of a venture began (Block and MacMillan 1985). The existence of initial customers, however, was no guarantee of future customers in sufficient numbers. JM feared additional customers would not exist, that the "phone will never ring."

As indicated earlier, all entrepreneurs did not confront comparable challenges in introducing their products across the supply and demand boundary. The greatest difficulty was faced by those who sought to introduce novel business concepts (embodied in suitable products) that required major customer education. This was either because of the slow and uncertain rate of customer learning, or the business concept was too early for the market, or it was poorly aligned with customer needs.

## Customer Feedback—Strategic and Operational

Linking with markets and evaluating and acting upon initial customer feedback are the concluding steps in venture creation. Introducing products to customers generated feedback for the entrepreneurs. Are sufficient numbers of customers interested in the product? Is the quality as desired? Do the features fulfill most contingencies in use? Is the product priced "right"?

Certain feedback affected the competitive stance of a venture, yet did not threaten its existence. Certain other feedback, however, threatened the business itself. To illustrate, in GB's words:

> What they (marketing) come up with are *revisions of existing equipment*, which is fine, for it is an improvement. There are typical ways of competing—we have features and two more and the price is 10 percent less. That has validity. But all of a sudden you come up with a *new approach to the problem*, and all of you folks are obsolete, no matter what features (emphasis added).

The need for "revisions of existing equipment" indicates feedback regarding the product features or product quality. But "new approach to the problem" is feedback affecting the business concept. It signals the emergence of substitutes in the market or radically different customer needs. Failure to act upon this feedback could undermine the very business.

A signal affecting the business concept is *strategic feedback* (Ansoff 1988; Child 1972; Maidique and Zirger 1984, 1985), because it addresses the entrepreneur's conception of the entire venture itself. If the difference between an entrepreneur's perception of customer needs

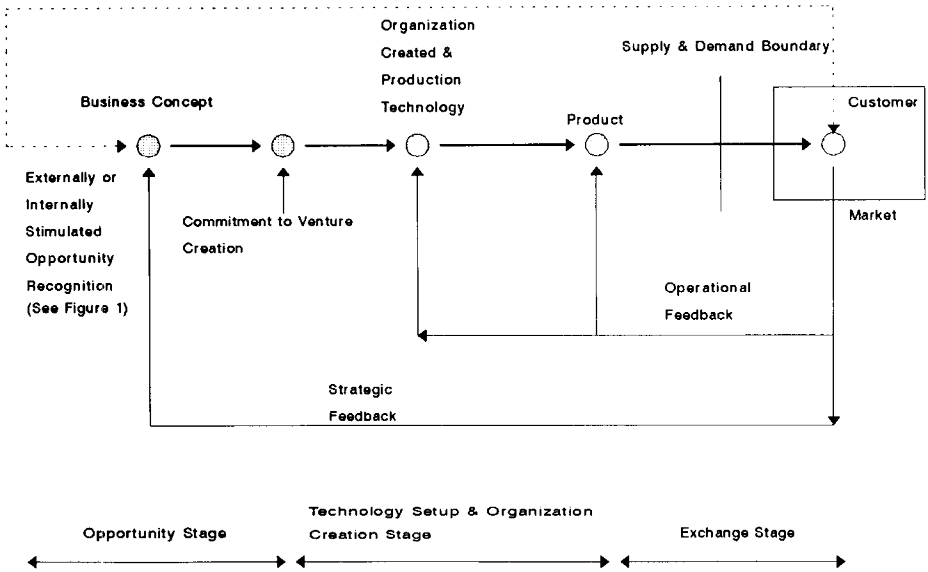


**FIGURE 2** Process model of entrepreneurial venture creation.

and actual customer needs is large, a major reorientation of the venture is indicated. Signals calling for improvements in quality, on the other hand, indicate the need for changes in production technology. If these signals are strong, a new generation of technology may be necessary. Signals for additional or altered features in products suggest the need for product changes. The latter two signals are *operating feedback* (Ansoff 1988) in that they require operational and tactical changes but do not directly threaten the validity of the business concept itself.

Feedback also appeared to have a *frequency* factor associated with it that called for quick responses (Child 1972). In GB's words:

> We were in a market where twenty new entrants came in every year and we had lots of competition. We were holding on, increasing and kept developing new products. We had to diversify into more product lines, which we did. Each is a major risk . . . to continue to grow at that fast rate would require continually going out and borrowing more funds . . . if you don't grow, someone else will, because you are *in a field that grows fast.*

## THREE PRINCIPAL STAGES IN ENTREPRENEURIAL FIRM CREATION

The conceptual categories and subprocesses discussed above are linked to generate the process model in Figure 2. For analytical purposes, and in order to elaborate upon the theoretical issues emerging from the model, it is divided into the three main stages, and an appropriate core variable is chosen to represent each. The stages are demarcated by natural transition points in the venture creation process.

Both internally and externally stimulated opportunity recognition sequences culminate in business concept identification. This is followed by the commitment to begin—a distinct step. The commitment to physical creation separates a hitherto invisible process from the subsequent visible one, and it is a natural transition point between two stages. Events in venture creation up to this point are therefore grouped into the opportunity stage, and business concept is chosen as the core variable to represent it.

After the commitment to physical creation, entrepreneurs garner resources and use them toward technology set-up, organization creation and marketing. This is the most visible stage in venture creation, and it concludes when a product ready for the customer is created for the first time. Because production technology is at the heart of this subprocess, it is chosen as the core variable representing the organization–creation and set-up of production technology stage.

Once an idea is transformed into a product and sold across the supply and demand boundary to customers for the first time, the entrepreneurial loop is complete. Customers directly evaluate the product, and generate feedback, both strategic and operational. The marketing efforts in venture creation as well as initial customer feedback and corrective action are all grouped into the exchange stage, and the product produced is chosen as the core variable to represent it.

## THEORETICAL ISSUES THAT EMERGE FROM THE MODEL

### Significance of the First Sale

The first sale is a significant *conceptual* event in new venture creation (Block and MacMillan 1985). This is because in making the first sale, the entrepreneur:

1. makes concrete the link between the business concept and the market that was recognized in the opportunity stage. The first sale vindicates the business concept.
2. establishes an exchange relationship with customers for the first time. Customer feedback, absent for the venture until now, becomes an input into a venture's future direction.
3. bridges the boundary between the supply side and the demand side.

The first sale is the last step in the physical creation of a venture, though venture creation has an iterative, conceptual component as described below.

### Conceptual Process

Opportunities are recognized and evaluated by entrepreneurs with respect to markets. This reciprocal correspondence between the opportunity stage and the market even *before* physical creation is shown in Figure 2 by the dashed line linking business concept and market. *After* physical creation, there is a more direct correspondence between the core variables and the market, as indicated by the feedback signals. And even *during* the physical creation process, the business concept may on occasion have to be modified according to the requirements of production technology, and product specifications may also require changes in production technology (Maidique and Zirger 1984). There is thus a reciprocal correspondence between the business concept and production technology and between the product and production technology.

Though the process in Figure 2 seems to follow a chronology, the feedback and reciprocal correspondence among the elements in venture creation suggests the existence of an iterative, *conceptual process* of venture creation, and of which the physical creation process is but a part. It may not be farfetched to say that the conceptual process is at least partially complete before the physical process begins, though refinements in it may continue well after a business is in existence.

Though the conceptual process prior to, during, and subsequent to physical creation cannot be directly measured, entrepreneurs value it immensely; it possibly explains their attachment to their businesses, and their reluctance to part with significant equity to venture capitalists (Beann, Schiffel, and Mogee 1975; Gladstone 1988). The financial valuation of a venture may be at odds with an entrepreneur's subjective perception of value based on the conceptual effort involved.

## Novelty and Entrepreneurial Content

When ventures in the set are viewed holistically with the aim of assessing the difficulty involved in venture creation, it is apparent that difficulties principally arise from the existence of high novelty at the core variables. This is because high novelty is generally resisted and the entrepreneur has to overcome this resistance (Schumpeter 1942; Abetti and Stuart 1989; Ansoff 1988; Meyer and Roberts 1986; Maidique 1980; Drucker 1985).

Though the general relationship of innovation to entrepreneurship is well known, it is seldom recognized that innovation in the entrepreneurial process can occur at all three core variables, and that entrepreneurial ventures can be innovative at none, one, or more variables. This suggests that the *entrepreneurial content* of a venture could be defined as a function of the novelty or discontinuity introduced, and therefore of the resistance overcome, at the three core variables.

## Distinguishing Among Ventures

The model in Figure 2 can be used to qualitatively distinguish one entrepreneurial venture from another based on the novelty at each core variable. A sizeable enterprise requiring large amounts of capital and employing numerous workers at start-up can have low novelty at each of the core variables. Though operationally complex, the uncertainty and the development effort with respect to the business concept, the production technology, or the product could be small. In contrast, a physically small venture may be based on complex conceptual underpinnings; it could have high novelty at more than one core variable.

This suggests that ventures could be classified based on comparable novelty embodied at core variables. Doing so may yield clustering of ventures having more in common than ventures classified along the traditional dimensions. Physically dissimilar businesses in different industries may have comparable novelty at core variables, and therefore may have more in common than ventures within the same industry. They may face comparable market creation and strategic issues. In contrast, businesses within the same industry, though physically similar, may face divergent strategic issues.

The literature surveyed for this research has found no attempt to classify ventures along these lines, though the effort at typology building as a component of theory building appears to be gaining momentum (Gartner, Mitchell, and Vesper 1989; Chrisman, Hofer, and Boulton 1988; Wortman 1987; Hambrick 1984).

## Common Entrepreneurial Dimensions

Entrepreneurship is a multidimensional phenomenon, and there is little agreement on common dimensions characterizing it (Gartner, Mitchell, and Vesper 1989). This process model suggests the use of business concept, production technology, and product as the preliminary set of core dimensions. These variables are sequentially encountered in the creation of all firms, deal only with the supply side of venture creation, and constitute a parsimonious set of variables. When each core variable is evaluated for novelty—either high, low or medium—it gives rise to a 3x3x3 cube of sufficient discriminatory power for preliminary analysis.

## CONCLUSIONS AND IMPLICATIONS

The process model developed in this study uncovers new ground as well as reaffirms some known theoretical issues in venture creation.

1. This study *identifies conceptual categories and stages* in the venture creation process based on empirical, grounded data.
2. It links the conceptual categories and stages to develop a *comprehensive, integrative process model* of new venture creation.
3. Despite its generality, the model has the ability to qualitatively *distinguish among entrepreneurial ventures* based on novelty introduced in the venture creation process at each of the core variables.
4. It further indicates that *novelty can be introduced at any of the three core variables*, and therefore, the adjective "innovative" needs to be cautiously applied to entrepreneurs and their ventures. Schumpeter's notion of innovation as the defining characteristic of entrepreneurship has thus been extended in this paper. Further it proposes that the *entrepreneurial content* of a venture is a function of novelty introduced during the venture creation process (Schumpeter 1934, 1942; Drucker 1985; Abetti and Stuart 1989).
5. The model suggests the existence of a *conceptual process of new venture creation* in addition to the more obvious physical creation process. Further, it suggests that this conceptual process is iterative, continues even after a venture is in existence, and is not linear or chronological.
6. It suggests *strategic gradations among entrepreneurial ventures*, and cautions against comparisons or groupings among firms along convenient dimensions.
7. Though the model recognizes critical entrepreneurial inputs at various stages in the venture creation process, and acknowledges the role of the entrepreneur as the prime mover of the process, it also suggests that in principle, *the entrepreneurial process can be viewed as a function that can be carried out by an organization.* Schumpeter did not rule out the absorption of the entrepreneurial function into mainstream business (Schumpeter 1942). In fact, internal corporate venturing may be viewed as bureaucratized entrepreneurship.
8. The model *reaffirms the existence of strategic and operation feedback* (Maidique and Zirger 1985) in venture creation. It concludes with existing research that the first sale is an important conceptual event in venture creation (Block and MacMillan 1985).

### Implications for Future Research

The model developed above suggests the following for future research:

1. Testing for the accuracy, completeness, and the relative weight of the concepts and subprocesses identified remains a future project. Though the data gathered are sufficient to develop the model, it is insufficient to address such issues as: are business concept development and marketing efforts greater when opportunities are internally stimulated rather than externally stimulated?
2. How similar or different are the processes of internal corporate venturing and entrepreneurship? Additional comparative research into the two processes is necessary to explore the common and contrasting features.
3. Future empirical research is needed to test whether the entrepreneurial content of a venture

is indeed proportional to the novelty introduced at the core variables as suggested here. Entrepreneurial content needs to be suitably operationalized, and novelty scales for each core variable need to be developed.

4. This paper indicates that typologies of entrepreneurial ventures can be developed using novelty at each of the core variables as the basis. Typologies so developed will aid the definition of taxonomic samples, and thereby enhance the external validity of research studies in entrepreneurship.
5. Internally and externally stimulated opportunity recognition, the three core variables, the role of novelty, the commitment to physical creation, the importance of the first sale, strategic and operational feedback and other conceptual categories, as well as the interrelationships among them, emerged in the process of analyzing transcript data. Some of these have been recognized previously in the literature as significant. The identification of these critical variables and subprocesses characterizing entrepreneurship points out the suitability, and indeed the advisability, of using ethnographic methodologies in emerging research areas such as entrepreneurship. Otherwise, conceptual categories and processes central to the emerging field will remain hidden, cross-sectional research will be limited to known concepts, and the advancement of knowledge in the field will remain stunted.

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